Joel Richardson · Curriculum Vitae

joelrichardson.au

Education

The University of Sydney — **Sydney, Australia**
Doctor of Philosophy (Mathematics) — *2026 - 2029*

- Thesis on Categorical Quantum Topology

Macquarie University — **Sydney, Australia**
Master of Research (Mathematics) — *2025 - 2025*

- Thesis on Monads and Tangent Categories
- 97.0 Thesis / 96.4 WAM

The University of Queensland — **Brisbane, Australia**
Bachelor of Computer Science (Honours) — *2023 - 2024*

- Thesis on Computer Algebra
- Dean's Commendation for Academic Excellence
- Honours Class I, 6.5/7 GPA

The University of Queensland — **Brisbane, Australia**
Bachelor of Mathematics/Bachelor of Computer Science — *2019 - 2022*

- Majors: Pure Mathematics / Programming Languages
- 6.3/7 GPA across honours level courses
- Reports on Coxeter Groups and Quivers

Employment

The University of Sydney — **Sydney, Australia**
Tutor, Mathematics — *July 2025 - Present*

The University of Queensland — **Brisbane, Australia**
Tutor, Mathematics and Computer Science — *February 2024 - November 2024*
Tutor, Mathematics — *February 2022 - November 2022*

Veitch Lister Consulting — **Brisbane, Australia**
Software Engineer — *September 2024 - January 2025*
Graduate Software Engineer — *November 2022 - February 2024*

See next page for talks and selected technical projects.

Talks

Australian Mathematical Society Conference — **Melbourne, Australia**
Talk on the tangent bundle monad — *December 2025*

Australian Category Seminar — **Sydney, Australia**
Talk on the tangent algebra comonad — *April 2025*

UQ Mathematics Student Society — **Brisbane, Australia**
Talk on Simplicial Sets — *August 2024*
Talk on Berlekamp's Algorithm — *April 2024*

Various High Schools — **(Various), Australia**
Taught braid groups at Groves Christian College — *November 2025*
Taught lambda calculus at Groves Christian College — *August 2024*
Taught lambda calculus at Trinity Bay State High School — *June 2024*

Selected Technical Projects

Focus: Formal systems and logic, programming languages, high-performance computing.

At Work

Compiler implementation (parsing, analysis, optimisation, etc.) — *2025*
Programming language design (type systems, syntax, etc.) — *2024*
Note: *language and compiler are currently deployed, with 10,000s of lines active in production.*

At University

Symbolic algebra system in Haskell — *Honours Thesis 2023*
Fast fluid simulation with CUDA — *HPC Project 2023*
First-order logic theorem verifier in Haskell — *(Prize Winning) UQCS Hackathon 2022*
Non-Euclidean rendering in C — *UQCS Hackathon 2020*

Side Projects

Formalising category theory in lean4 — *2025*
Winning games with computation: wordle, snatch, articulate — *2021*
Neural networks in C — *2021*
Software rendering pipelines in Java and C — *2020*
Designed and built a CPU inside Minecraft — *2018*
Autoencoder neural network in Java — *2017*